

## Final Project Instructions – Advanced Database Systems

### Objective:

Each student must individually design and develop a unique mini database system that demonstrates mastery of database design, advanced SQL concepts, and GUI-based CRUD operations.

Your mini system should be small yet realistic, capable of functioning as a standalone application that integrates both a graphical user interface (GUI) and a fully functional database backend through API implementation.

### \*\*\* Project Requirements

#### 1. Individual Work

Each student must propose and develop their own unique mini system (no duplicates).

The system must be realistic, practical, and manageable within the scope of this course.

#### 2. Functional Requirements

The system must include CRUD operations (Create, Read, Update, Delete) for at least four (4) related tables.

The GUI should allow users to perform CRUD operations intuitively through forms, tables, and buttons.

Include sample data with at least 1,000–2,000 records per table to test performance and indexing efficiency.

#### 3. Advanced SQL Features

Your database must demonstrate the following advanced SQL features, each integrated logically into your system's operation:

- ✓ Trigger
- ✓ Stored Function
- ✓ Stored Procedure
- ✓ View
- ✓ Index
- ✓ Query with Subquery

(All features must serve a clear functional purpose and not exist as isolated examples.)

#### 4. Graphical User Interface (GUI)

The GUI may be developed using PHP/MySQL, Python (Tkinter or Flask), Java (Swing/JavaFX), or any environment approved by your instructor.

The GUI must include:

Add, Edit, Delete, and View options

Proper input validation and error handling

Real-time data display retrieved from the database

A user-friendly and organized layout

Notification or confirmation messages after performing CRUD actions.

Also add a dashboard for data visualization using advance queries.

#### 5. Database

Use Microsoft SQL Server 2019 (or the latest version available) as your primary database management system. Implement API-based communication between your application and the SQL Server database for secure and modular data access.

Ensure all SQL scripts (schema, data insertion, and feature creation) are properly documented and executable.

#### 6. Application Environment

The system may run locally on your machine (no online deployment required).

The application should demonstrate database connectivity, functionality, and performance through your API layer.

### \*\* Suggested Development Tools

Database: Microsoft SQL Server 2019 - latest version

API Framework: PHP, Python Flask, or Node.js (depending on language)

Frontend (GUI): PHP (HTML/CSS/JS), Tkinter, JavaFX, or similar